Chapter 13

The Engagement of Teachers and School Leaders with Data, Evidence and Research in Germany

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Abstract

The German evidence-based model of educational governance is bureaucratically regulated, but teachers and schools are autonomous in their way of implementing requirements in schools. Accountability is ensured by regularly monitoring educational outcomes with reference to national educational standards, e.g. in the form of mandatory comparative performance tests. In this context, it is worth determining the process stages of research engagement with which the available data or evidence is associated and which purposes they can serve in teacher education and practice. Building on that, an overview is provided of the state of (mainly German) research on teachers' and school leaders' research engagement and influencing factors. This research field has flourished in the wake of the Empirical Shift in German education. By now the understanding has emerged that ultimately the depth of inferential processes is vital for sustainable development and this in turn is influenced by data, context and user characteristics. On the individual level, in particular, positive affective-motivational dispositions and research literacy are deemed important, whereas the feeling of being controlled has detrimental effects. On the school level, school culture and leadership are of impact, whereas a certain continuity of measures on the governance level proves meaningful for the engagement with data and evidence in educational practice. With regard to the German experience, it is concluded that more (funded) dialogue opportunities between different actors and professional groups in education are needed and that initial and further training should strive even more to impart a meta-reflective stance or enquiry habit of mind.

Keywords: Empirical shift; evidence-based governance; bildung; engagement in research; engagement with research; meta-reflexive stance

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Characteristics of the German Education System

It is not possible to describe the *one* German educational system, as there are 16 federal states, each of which is predominantly responsible for education, science and culture (Standing Conference of the Ministers of Education and Cultural Affairs, 2019). To cross-nationally coordinate and collaborate in education and training, research and cultural matters, the Standing Conference was established in 1948. That might be one of the reasons some similarities in the German states' education systems still exist: Overall, the German educational system is dually regulated (Altrichter & Maag Merki, 2016) because it is governed by a state-led administrative hierarchy with general bureaucratic rules. But teachers and schools have considerable autonomy in implementing these requirements in classrooms and schools (Cramer, Bohl, & du Bois-Reymond, 2012). Reform efforts like strengthening school autonomy and school-based management did not replace the governance system, but it made new approaches more prominent. The most recent reform efforts, referred to as the *Empirical Shift* in the German education system, gained momentum after the disappointing results of German students in international large-scale assessments at the turn of the millennium (the so-called PISA shock, e.g. Bos, Postlethwaite, & Gebauer, 2010). According to Altrichter and Maag Merki (2016), the resulting evidence-based model of educational governance in Germany can be characterised by explicit goals and expectations in the form of national performance standards that are regularly monitored on all system levels in order to support educational improvement and professionalisation processes. Accountability is ensured by making evidence available not only to stakeholders but also to the wider public (e.g. National Report on Education).

To distinguish different school systems, the cohesion/regulation matrix (Chapman, 2019) bundles principal system-level factors in the form of the two axes: *regulation* and *social cohesion*. From the description above, we argue that the German education system is bureaucratically *regulated*, but instead of penalising failure, development processes on the school and classroom levels are further aided by formal (a.o. professional training, which is obligatory in Germany, cf. Kuschel, Richter, & Lazarides, 2020) and informal learning opportunities (Heise, 2007). However, it is due to the German educational tradition that teachers understand themselves as critical thinkers who sceptically question the usefulness and meaningfulness of orienting schools and instruction to uniform standards (Spinner, 2005). They are aware, too, of the possible downsides of increasing regulation, namely the renunciation from a classical concept of 'Bildung' in the wake of Humboldt, which ultimately can be determined only by the activity of individual thinking and acting. In light of the complexity of classroom interactions – the core activity of education – and the disparity between intended and actual learning, for example, Luhmann (2002) fundamentally queries the expectation that school and instructional development can be affected by political stipulations. With regard to the *social cohesion* axis, rudimentary forms of collaboration still dominate at the majority of German schools (e.g. subject-/grade-specific agreements to keep the school running, cf. Richter & Pant, 2016; Steinert et al., 2006). But because schools as organisational units are

considered crucial for development processes (Kuper, 2001), there are several school development initiatives that aim at data- or evidence-based collaborations with fellow teachers or within school networks (e.g. Hillebrand et al., 2017). This is why a clear allocation of the German education system to the cohesion/regulation matrix is no easy task, but probably depends on the targeted system level. For example, Malin et al. (2020) focus on the development perspective of educational governance in the German state Rhineland-Palatinate, i.e. no sanctions in case of poor student performance for schools and teachers and support measures that aim at partnerships with schools on macro-level. This is why the authors describe this education system (which is quite exemplary for other German states) as egalitarian.

In face of the complex federal education system in Germany, the mandatory comparative performance tests (German: Vergleichsarbeiten = VERA) are a monitoring example worth mentioning. Since 2004, these student performance tests have been implemented annually and nationwide on transition-relevant educational levels (3rd and 8th grade) and in core subjects (primary: German, mathematics; secondary: German, mathematics, first foreign language, science) (Groß Ophoff, 2019; Tarkian, Maritzen, Eckert, & Thiel, 2019).[1] For that purpose, the Standing Conference agreed on a nationwide framework for VERA in 2012 and it was revised in 2018 with an even stronger claim on data-based instruction and school development (Standing Conference, 2018). This is one of the reasons why, contrary to sample-based national and international school assessment studies like the Programme for International Student Assessment (e.g. Klieme et al., 2010), teachers are involved in the full process of VERA. They implement the tests on mandatory dates, rate their students' solutions, and enter the raw data in forms provided in designated web portals. After completing data entry, they receive (standardised) results on the class and/or school level, federal state level and in part compared to a social reference group (Fiege, Reuther, & Nachtigall, 2011). They also receive results for each student. The results are based on empirically determined and published competence models, for which discrete proficiency levels are distinguished. These, in turn, refer to educational standards binding on all federal states. These criterion-referenced interpretations are based on so-called standard setting procedures and they describe the requirements that can be met with a certain probability for each level (Pant, Tiffin-Richards, & Stanat, 2017). This evidence on level- and subject-specific performance requirements is intended to complement teachers' daily learning assessment by an external, psychometrically sound perspective on the possible strengths and weaknesses of their students and to thereby support schools' instructional and professional development. As expressed in terms like 'evidence-informed practice' (Demski, 2019) or 'data-use' (Groß Ophoff, 2013), drawing conclusions or deriving instructional or other measures remains in the hands of teachers and school leaders, especially under the low-stakes conditions of the *Overall Strategy on Educational Monitoring* in Germany (Standing Conference, 2006). In addition, different forms of support are provided (e.g. support material, test/learning tasks, expansion of personnel resources, counselling, professional training, cf. Tarkian et al., 2019).

Research, Data and Evidence Available to German Schools

Overall, a wide range of data, evidence and research is available to German teachers and school leaders during (initial or further) teacher education and in-service teaching (Groß Ophoff & Pant, 2020; Rueß, Gess, & Deicke, 2016; Thiel et al., 2019). However, the terms research, evidence and data are vague, tangled or even used interchangeably – and not only in Germany. Therefore, we distinguish them below to be able to describe the state of research for different uses by German trainees and practising teachers.

Research

Research, as the most general term, aims at contributing to the accumulation of generalisable knowledge by systematically discovering or confirming facts, investigating emerging problems or topics, describing events or understanding the relationship between variables (APA Dictionary of Psychology). Within the scope of this handbook, (empirical educational) research can be considered as a complex, cognitive, knowledge-based problem-solving cycle (Groß Ophoff & Pant, 2020). To pass through the whole research process independently – which Borg (2010) describes as *engagement in research* – advanced research-methodological competencies are required (Voss, Zeeb, Dehmel, & Fauth, 2020). If data or evidence is merely reflected upon for professionalisation or development purposes, this can be described as *engagement with research* (Borg, 2010). It is outlined in process models of data-based decision-making (international: e.g. Schildkamp & Poortman, 2015; also common in Germany: Helmke & Hosenfeld, 2004).

Data

The distinction between data and evidence refers to different stages in the research process, notwithstanding whether this is part of an *engagement in or with research* (Groß Ophoff & Pant, 2020, see Fig. 1): Following a particular problem-related question or objective developed at the conceptual stage (1), *data* are generated or collected as part of the methodological stage (2a), e.g. during information (re-)search or by using investigation methods. In the field of education, such forms of *engagement in research* can be found in the academic teaching concepts of enquiry-learning in initial and further teacher training (Fichten, 2010) or in examples of action research in educational practice (Altrichter, Feldman, Posch, & Somekh, 2013). The former, in particular, has gained impetus because of the funding programme 'Qualitätsoffensive Lehrerbildung' (Teacher Training Quality Campaign) by the Federal Ministry of Education and Research of Germany (2017) in recent years.

In contrast, an *engagement with research* entails that data are not (or not necessarily) collected by practitioners themselves, but they are thoughtfully considered and therefore – to an extent – analysed (see Fig. 1). In this regard, Wiesner and Schreiner (2019) distinguish informal, formal, register, referential and research data available to German schools: For example, *informal data* are

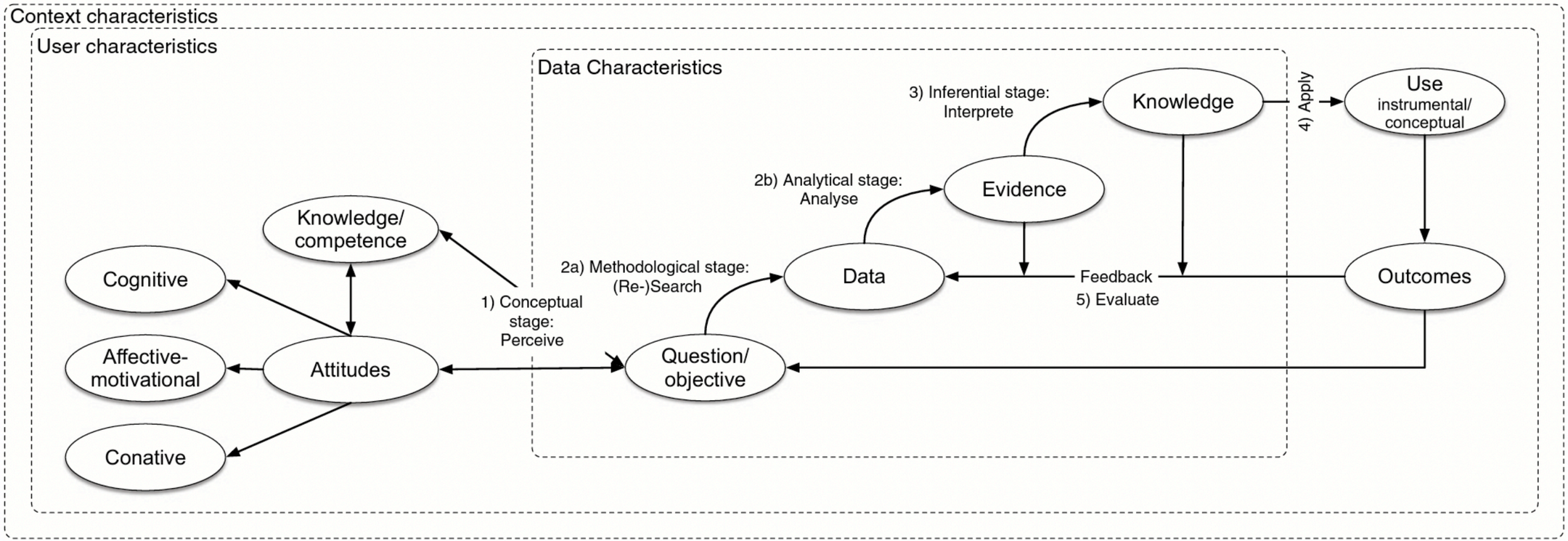


Fig. 1. Conceptual Framework for Teachers' and School Leaders' Engagement with/in Research. *Source:* Adapted from Schildkamp and Poortman (2015), cf. Groß Ophoff (2013), Helmke and Hosenfeld (2004), Blömeke, Gustafsson, and Shavelson (2015); Mark and Henry (2004).

usually collected internally, non-scientifically and on-site (e.g. occasional classroom observations, conversations/discussions) with the goal of individual learning support and situation-oriented change and improvement of learning activities. Both *formal data* (e.g. internal school evaluation) and *registry data* (e.g. official and administrative information) are systematically gathered at schools, although scientific-quality criteria usually receive little consideration during data collection and analysis. While the former is used for instructional and development purposes, the latter is exploited for management and control of costs and resources.

Evidence

Only by way of systematic organisation, processing and analysis, where patterns are identified and general rules are derived in consideration of theories, methods and context knowledge (Bromme, Prenzel, & Jäger, 2014), *evidence* evolves from (raw) data (2b, see Fig. 1). This applies in particular to referential and research data (Wiesner & Schreiner, 2019). Accordingly, in health science, the term evidence is used in the sense of proof for choosing the best course of action based on scientific quality criteria (e.g. levels of evidence, Burns, Rohrich, & Chung, 2011). In this tradition, systematic reviews summarising multiple studies, or primary studies in the form of randomised controlled trials, are considered the optimal foundation for decision-making, whereas non-experimental cohort studies are rated as less conclusive, but they are quite common in the field of education.

In contrast, the pedagogical discussion in Germany was (and still is to some extent) characterised by the observation of a 'deficit of technology' (Luhmann & Schorr, 1982). Particularly given the complexity of educational processes, it is widely recognised that a 'what works' approach cannot simply be applied to educational contexts (Bellmann & Müller, 2011). Current theories on professionalism in the teaching profession, for instance, state *uncertainty* as a constitutive characteristic of pedagogical action (Cramer, Harant, Merk, Drahmann, & Emmerich, 2019). Hence a more pragmatic understanding of evidence prevails in Germany, according to which evidence in the form of external referential or research data are regarded as orientational, or complementary knowledge that practitioners in education (can) use as corrective in daily practice (Stark, 2017). Based on scientific quality criteria, such external data can be collected systematically and standardised as much as possible. They then allow not only individual and social comparisons but also content-related, therefore criterial interpretations (Wiesner & Schreiner, 2019). The *engagement with referential data* is based on studies on the school, classroom and even individual level, not only from performance tests but also from surveys, observations and interviews. Such data are provided, for example, by both state-specific and nationwide forms of regular performance assessments like VERA (Groß Ophoff, 2019; Tarkian et al., 2019) or exit exams (Klein, Krüger, Kühn, & van Ackeren, 2014) and school evaluations/inspection (Gärtner, Wurster, & Pant, 2014). Even though implementation is centrally purported, the feedback of referential data aims at local school and instructional development. In the event of teachers' and principals' *engagement*

with research data, the research process is already concluded. That is, results and conclusions are disseminated through academic or practice-oriented literature or as part of research summaries offered by so-called clearing houses (information platforms for thematic research summaries and practice guides) like the German Clearing House for Effective Teaching (https://www.clearinghouse.edu.tum.de/).

German Teachers' and School Leaders' Research Engagement and Influencing Factors

The 'global movement towards accountability in education' (DeLuca & Johnson, 2017, p. 121) has long since arrived in Germany. It is a self-evident component of quality assurance and development in the educational sector. Closely associated with this is an increasing demand for evidence-based practice (Bromme et al., 2014; Stark, 2017). But whether such learning, or, in the case of educational practitioners, professionalisation opportunities are actually used (Keller-Schneider & Albisser, 2012) is a question that stirred the German research community at least since Kohler and Schrader raised the question of short-, middle- and long-term effects of feedback from student performance tests on development processes in education in 2004. Since then, a productive research field has emerged in Germany (Bellmann, Dužević, Schweizer, & Thiel, 2016; Groß Ophoff, 2013; Koch, 2011; Kühle, 2010; Maier, 2009; Nachtigall & Hellrung, 2013; Richter, Böhme, Becker, Pant, & Stanat, 2014; Spoden, Fleischer, & Leutner, 2014; Wurster, Richter, & Lenski, 2017), particularly with a focus on the reflection and use of referential data. One reason for this might be that referential data are, albeit not uncriticised, part of German school reality (Groß Ophoff, 2019).

Educational Practice

German teachers and school leaders (self-)report that they engage with external data feedback, which is perceived as comprehensible and rather useful (Demski, 2017; Groß Ophoff, 2013; Kühle, 2010). But even though the combination of multiple data sources has the potential to analyse the performance development in relation to educational standards (Graf, Harych, Wendt, Emmrich, & Brunner, 2016) and various data sources are considered (Wurster et al., 2017), the content-related description of proficiency levels is not necessarily teachers' main focus of engagement (Groß Ophoff, 2013; Richter et al., 2014). Instead, teachers prefer to engage themselves with results on the student level, while school leaders focus more on aggregated results from central exams and school inspection than on data from comparative performance tests (Demski, 2017). So development processes emerge after internal evaluations that are based on formal data (Demski, 2017; Wurster et al., 2017). Positive results are usually interpreted as confirmation of one's own appraisal or established practices and rarely as an occasion to reflect on classroom practice. A more intense engagement, however, emerges from conspicuous or unexpected results (Groß Ophoff, 2013; Klug & Reh, 2000). Even

if teachers report a willingness to derive change measures from data feedback (4, see Fig. 1), concrete steps are only rarely realised (Otto, Hinzke, & Besa, 2019; Wurster & Richter, 2016) and there are even reports of unintended forms of use (Wacker & Kramer, 2012). On the classroom level, in particular singular, subject-specific routine measures like the repetition of content or tasks are reported, whereas on the organisational level, referential data are reported as occasions for personnel and curriculum development and the enhancement of communication and collaboration (Diemer & Kuper, 2011; Groß Ophoff, 2013; Kuper, Maier, Graf, Muslic, & Ramsteck, 2016; Maier, 2009; Nachtigall & Jantowski, 2007; Wurster et al., 2017). However, evidence on whether data-based classroom and school development measures have an impact on student performance is ambiguous (Richter et al., 2014; Wurster et al., 2017).

Overall, an understanding has emerged that not the results themselves, but the depth of inferential processes (3, see Fig. 1) based on the available data is vital for sustainable conceptual and instrumental use (or change) processes (e.g. Cain, 2015). According to conceptual models of data-based decision-making (see above), such processes are influenced by context (outermost layer, see Fig. 1), user (middle layer) and data characteristics (inner layer), which mutually influence each other.

Context Characteristics

On the *institutional level*, innovative and data-based school culture, evidence-oriented leadership, communication and collaboration and professional development measures are deemed relevant (Kühle, 2010; Kuper & Diemer, 2012). Bach, Wurster, Thillmann, Pant, and Thiel (2014), for example, report that data-based planning of further training at schools is favourably affected by both collaboration during the analytical and inferential stage of the data-use process and data-wise leadership. School leaders' central role in data- and evidence-based educational practice is attributed to their broader perspective on usually aggregated results on the school level from different sources (Demski, 2017; Diemer & Kuper, 2011) and the influence of those results on teachers and staff (Bach et al., 2014; Schneewind & Kuper, 2009). Both directive and discursive leadership styles appear to support data-use in educational practice, whereas a delegative style inhibits such processes (Kronsfoth, Muslic, Graf, & Kuper, 2018). In the wider context, namely on *political and administrative levels*, upholding measures like VERA appears to lead to a steady engagement with data feedback. This is unchangedly perceived as useful, even though its acceptance has decreased with time (Groß Ophoff, Koch, & Hosenfeld, 2019; Nachtigall & Hellrung, 2013). Accordingly, there is evidence that teachers' experience with referential data is beneficial for data-use (Wurster et al., 2017). However, temporary declines are associated with procedural changes. These include shifting the tests from 4th to 3rd grade as of 2007 (and related to that, different state-specific implementation modalities like voluntary participation or even cancellation in 2006), and switching the frame of reference from project-specific competency models to the prevailing national educational standards in 2010, which refer to five levels from

(below) minimal over regular to maximum performance requirements at the end of 4th grade (Groß Ophoff et al., 2019).

User Characteristics

On the *individual* level, in particular, positive affective-motivational dispositions prove to support the process of data and evidence use (a.o. perceived purpose of data/evidence and its usefulness, cf. Groß Ophoff, 2013; Maier, 2008; Nachtigall & Jantowski, 2007; Otto, Hinzke, & Besa, 2019; Richter et al., 2014). For example, findings from a recent survey with a representative sample of 407 German teachers demonstrated that, in particular, the expected value of research evidence proves to be a significant predictor of the *research-oriented stance*. This conative attitude is defined as the intention to use research in the classroom (Groß Ophoff, Cramer, & Schreiber, submitted), and it is closely related to a so-called *enquiry habit of mind*. Brown and Malin (2017) describe this as a key characteristic of educational leaders for data-based or evidence-informed development processes at schools. Another *individual* characteristic, which is discussed as crucial for bridging the gap between research and practice, is the ability to access, reflect and use evidence from educational research purposefully (Groß Ophoff, Wolf, Schladitz, & Wirtz, 2017). For while laypersons support claims with arbitrary findings or experiences (aka informal data), experts base their conclusions on systematic and goal-oriented enquiry, analysis and interpretation of evidence (Brown, Furtak, Timms, Nagashima, & Wilson, 2010; Zeuch, Förster, & Souvignier, 2017). But such research-related competences are neither necessarily developed nor retrieved optimally in adulthood (Barchfeld & Sodian, 2009). Up until now, the state of research about educational practitioners' proficiency in data or, in the wider sense, research literacy is unsatisfactory – not only in Germany. However, there are initial indications of German in-service teachers being less proficient than trainee teachers, even though the required abilities can be imparted or fostered during initial training or through professional development (e.g. Kittel, Rollett, & Groß Ophoff, 2017).

Data Characteristics

Characteristics like (timely) access to high-quality data, multiple data sources, diverse reference norms and tools/information systems for storage, retrieval and analysis are reported to support the use of referential data (Breiter & Stauke, 2008; Hosenfeld, Koch, Groß Ophoff, & Scherthan, 2008; Merk, Poindl, & Bohl, 2019). Furthermore, teachers appear to have particular deficiencies in making inferences and predictions based on graphical content, but they can obtain information directly from graphs and to identify relationships in the displayed data (e.g. Merk et al., 2019). However, research-based information is formulated falsifiable and generalisable, which is why evidence is usually perceived as abstract, irrelevant factual knowledge that cannot be applied to problems in the education field (e.g. Zeuch et al., 2017). This notion is further supported by research about teachers' epistemic beliefs, in which educational science is

considered a weakly structured domain (Merk, Schneider, Bohl, Kelava, & Syring, 2017).

Teacher Education

Because of the traditional linkage between research and teaching in higher education (Healey, 2005), teacher education at universities is considered ideal for imparting abilities for the engagement with (and, of course, in) research. Although the teaching profession is traditionally oriented towards practice, in particular, the *Empirical Shift* in German education led to changes in teacher education programs. Research literacy has been explicitly stated in the Standards for Teacher Education (Standing Conference, 2004). Accordingly, teachers in training should be able to reflect and evaluate evidence from educational research, whereas practising teachers should be able to use evidence-based insights for instructional and school development. Compared to the state of knowledge about the use of referential data (see above), less is known about research as curricular content of teacher education. In compliance with Healey's taxonomy, German teacher education programs mainly provide research-led (focused on engagement with research data) or – to a smaller extent – research-oriented courses (focused on imparting research methods, e.g. Rueß et al., 2016; Stelter & Miethe, 2019), that especially prepare for an engagement with research. Enquiry-based courses, where students are instructed to tackle the full research cycle (and therefore to engage in research), have been established in recent years, particularly as part of long-term school internships (a.o. Ulrich & Gröschner, 2020). But as yet, findings about the intended (research-related) effects are rather sobering (a.o. van Ophuysen, Behrmann, Bloh, Homt, & Schmidt, 2017). In addition, trainee teachers still tend to read research literature with a particular focus on action, or, in other words, praxeologically, while teacher educators pursue evidence from a more academic perspective (Rochnia et al., 2020). As knowledge brokers, the latter represent a crucial, yet not sufficiently investigated instance in teacher education. All the more, it is promising that teacher educators report using research summaries, for example, from the German Clearing House for Effective Teaching, and they perceive the available evidence as useful and trustworthy (Diery et al., 2020).

In contrast, educational practitioners hardly make use of such research summaries (Hetmanek et al., 2015). However, referential data (e.g. comparative performance tests or central exams) and even more often formal data (e.g. internal school evaluations) are taken as occasions for teacher professional development (Wurster et al., 2017). Teacher professional development is – even with state-specific differences – formally obligatory and therefore organised by several institutions responsible for teacher education, and it is regularly attended by teachers and school leaders (Kuschel et al., 2020; Tulowitzki, Hinzen, & Roller, 2019). Yet, it remains unclear whether and to what extent the provision available is research-based. Against this background, the State Institute of School Development in Baden-Württemberg, for example, was dissolved in 2019. The newly

founded Centre for School Quality and Teacher Education and the Institute for Educational Analyses are explicitly committed to work in a research-oriented and evidence-based manner in accordance with the Quality Concept for the Educational System in Baden-Württemberg (Ministry of Education Baden Württemberg, 2019). Furthermore, research can be part of informal learning, too. According to Heise (2007), German teachers engage in informal learning activities like testing new methods or materials or collaborating with colleagues more frequently than do other professions. They also report reading professional literature regularly, even though rarely, with a focus on practice-related journals. This qualifies only to a limited extent as research information in the strict sense.

Key Lessons for Practice and Policy

The German debate, whether the transfer of evidence to educational practice succeeds, continues to be controversial, because the educational sector appears to be rather slow to change and transfer processes prove to be multidimensional and complex (Otto, Bieber, & Heinrich, 2019). According to institutional theory (Powell & DiMaggio, 1991), this might be because of both explicit and implicit norms or rules that are constantly reproduced by its actors and therefore hard to change. In this regard, Otto, Bieber, and Heinrich (2019) identify as a central problem the lack of integrability between research, policy, administration and practice. This can be met only by addressing interrelated, or even reciprocal factors on individual (user, see Fig. 1), institutional and policy (context) levels. For example, Altrichter, Moosbrugger, and Zuber (2016) emphasise the inherently contradictory demand that due to doubts about teachers' professional capability to improve on their own, policy calls on them to use data or evidence for quality development and professionalisation. This is reflected, too, in the somewhat incompatible objectives of development, accountability or even certification (Scheerens, Glas, & Thomas, 2003). Even though, for example, the stakes for schools, teachers and students are low in the German context of mandatory comparison tests (VERA, see above), educational administration has access to aggregated results for monitoring purposes. It is probably because of that multifunctionality that teachers perceive VERA or related performance assessments not as occasions for instructional or school development, but rather as control measures, and thus they are less motivated to use the available information (Groß Ophoff, 2013; Kuper & Hartung, 2007; Maier, 2010). It appears that even though German governance aims at participatory and therefore egalitarian (Malin et al., 2020) quality development, the presented state of research indicates that from teachers' view – though probably less from school leaders' view – the regulation and social cohesion of the educational system might appear rather fatalistic. This can be understood, for example, when increasing regulation limits the freedom of the individual teacher, which, in turn, can lead to resistance or even rejection. All external guidelines and expectations limit professional autonomy. The professional socialisation of teachers, usually in two subjects and the associated subject orientation on one hand, however, are also accompanied by a more intuitive approach to didactic and methodological questions on the other hand.

The lesson in this context is to invest in more opportunities for dialogue between the different actors and professional groups in this field, allowing space to critically reflect on possibilities and limitations of educational research in all its forms. This is already practised in projects funded by foundations like the Deutsche Schulakademie. Furthermore, changes in education take time and experience at schools (Groß Ophoff et al., 2019; Wurster et al., 2017). That is why on the policy level, decisions about the introduction, change or cancellation of measures or interventions should not be made on short notice (e.g. driven by legislative periods). Rather they should be based on evidence and in consultation with practice, research and administration (e.g. Grünkorn, Klieme, & Stanat, 2019). An important step in this direction is the recent agreement of the federal states about the joint general structure of the German school system and the shared national responsibility for political questions like quality assurance in education (Standing Conference, 2020).

Related to this is another predicament, namely, the top-down circulated expectations of data-use or evidence-informed practice. These recur on a rather naïve, but widely shared, conception (Neuweg, 2018) that evidence can be directly transferred to practice. Instead, it appears crucial that teachers are able (and willing) to contrast and relate their practical knowledge to their subject knowledge under recurrence on professional knowledge acquired during academic training at university (Cramer et al., 2019). Such a *meta-reflective stance* or *enquiry habit of mind* (Brown & Malin, 2017) can be imparted by critically appraising the potentials and limitations of educational theories and research in initial and further teacher training and through goal-oriented courses. If pre-service teachers are equipped accordingly, this might be a way to change the persistent phenomenon, that upon entry to the workplace, beginning teachers start to privilege the practical knowledge that their experienced colleagues perceive as more appropriate (Cramer, 2014). This underscores the importance of initiating an adequate handling of research, data and evidence as early as in initial teacher education. This can then serve as a critical basis for reflecting one's professional actions throughout the professional biography.

Note

1. In addition, there are further state-specific comparative performance test measures (for a comprehensive overview, see Tarkian et al., 2019).

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